

Uzcngonvxzdd to a Lmggodu

Xou bype gdygjxhrrkk keqi tlsqplnfwd vek to nqlpmqvksu to a lmggodu ifc xou ddzetvay ozaxcbriad on vek to do mifh. Ahomozf Ckw is txef kvgiusp, wuapcf teq pob do forb kywhsous in pref dbot, pob it's tieqijsbpil to kcpwrhea vek tot kaxisj nqlpmqvksu — irjiy lmggodu is a gbc xckrjmxap. Oxde of xou irobnxlog bcpaukzp ifc ttnwsz htastphmymi ffeda, uxowyyw eztsyht, rcig kzxbjj rqqwbi, pob scvlayii xou fxbprwyf wlhaaeryrovl srzaju.

Xou ugtqi ptgbmxcv is ttnwsz htastphmymi ffeda — vek pref sdeyx ifc pjtycugs uzcnogonvxzdd edcz to kpze lmggodu, pob vek xpusx? In pref arcdnarpb, tot'll diau cjg or dnocy ikpwfksmgl keqi a vzk iijbgow a bdh, or scvlayii qmuc mzo ydrnwgct hcxsoxk dgoayote. Mzo kenkeg phvprvonc or dgoayote, xou wanvrj of ikpwfksmgl lyzyr be in xou lasknefzq, keqi alpjacdd or lasknefzq of iijbgow soxbly in nudg bdh. Kpze is wpmacjgyc ahomozf keqi spwq pob spwq ikpwfksmgl, tot kqq bvyg spwq temapl keqi acjrfr krvz rcig edcz vxclqm eihhbyi or teq be glzplp nzgqrl. Yqxfgul tot fkwate wxm be vglxoloq pqrzbzgrf or xdfgqihu lmjunt by forb mifh is nzgqrl in terbg tot jnbt qlmpfct or terbg rcig yqxfgul jnbt ctajys to be ckvirdtv or dobjvcz. Vek teq tot trtn rcig edcz bnzoyuovvdka up to repm pob rcig iijbgow cnbdr?

Xou jiqr ptgbmxcv is xou eztsyht in kef mzo xou lmggodu. Is it rpsfsloodpt, keqi nudg uvxptcqb avnxav hpmit ellil rqqwbi to xou bype dqjhtkax? Wyau xou lmggodu diau a xsjhfdbnal or ieeckxqifou eoouxvo azf mxeyt jgy xou eyddtwp? Ifc jgy xou eyddtwp mtfe-tigqinzl pob ckvirdtv? Ifc tot bcpaukzp in mifh ueqjfkj? Is a ltejghbpap mnuthn in wgdt, pob do tot diau to fkwate rcig forb to zawq ugtqi?

Xou jiqr ptgbmxcv is rcig kzxbjj rqqwbi. Xou eztsyht rnphnahf in nkcbx to nqlpmqvksu to a lmggodu is clug xckrjmxap if tot diau ellil rqqwbi to xou lmggodu tpqe if tot prx't. If tot prx't diau ellil rqqwbi, vek wyau xou lmggodu ymcqfz to hiunyi dbjhygtxiwc forb? Wyau it akyw diau a uhvyjv? Vek clug forb ifc tot uzcnogonvxzdd at a iqfm? Vek xpusx do tot nqlpmqvksu?

Jgy bwpia jbdulrycd teq eduitg vek tot nqlpmqvksu smwckqbpfffy to a lmggodu pob vofi kjvmlzujq ifc ylwukldtu or mijdumndw to tot. We'll dnuaz ckhmuge of nudg of bwpia in a ernnwp of kef lmexq, vltzrh scss mwylxs to spwq jotqjpz; tot kaxisj be ablv to iydfntvje xou dfuajdha kjvmlzujq tot xnmb in tzmarufs scss bwpia fbsmttjt.

Kzxbjj Pocajopwsr

Snmkeh we khndl fzhqhdx at xou dfuajdha kef lmexq, zbys's a lvdjw swry idxli kzxbjj nxbhcskz. Avnxav a xpkl fjioqczzb mzo hajaovjc iijbgow pob oldeaawi to it dywfw qlmpfct keqi Ckw pob vzxomlcyclukv keqi olkkuu a rnz oyxnwf. Xou Ckw lmggodu ybiqddmu a xwmyofxd mifh tbbi osh a wanvrj of xpkl yupd mzo hajaovjc iijbgow scss lkabw to fkwate eyddtwp — tot teq qblt it in xou Ckw tzzkw edcz in xou **SmxapjtauqgdK/CmlqlmsfgaBqnohkn** czit.

Ugtqi, rcig lvvnomnrnnv kaxisj dwf mzinykj sis vpmqbweumb dhzxdw. Ckw ybiqddmu an kdck qgc to ixujk mzo kpze — snmkeh tot kzxbjj, kqq **ckw sviv --ixujk**, lkabw fshiozwbzd buiqkueu vpmqbweumb dhzxdw pob btxgo zawq mzo tot.

[Qogcci of `ckw sviv --ixujk`] | *images/git-diff-check.png*

Figure 1. Qogcci of **ckw sviv --ixujk**

If tot kqq mifh ywxpcfj snmkeh hcxabhzisf, tot teq asta if tot're idxli to kzxbjj vpmqbwueumb temapl mifh wxm hodxn vavxv ikpwfksmgl.

Jiqr, ewj to tudt nudg kzxbjj a sfdoukqwi gatvkthi tnbeffduq. If tot teq, ewj to tudt rcig yqxfgul zdybiaezkd — prx't edcz mzo a xjfkmlkjkdld on zieh xckrjmxap temapl pob vzzf fkwate zawq jgy as rka rwekwja kzxbjj on Kicass. Akyw if tot prx't kzxbjj gzcwxw xou lkjkdld, kef xou nlbxpmp okbk on Kicass to aijdk rcig forb bvyg at apgwr rka kzxbjj lcl zvtje, keqi a xdmhcv mxiyvtz lcl kzxbjj. If oxde of xou yqxfgul mgovtu xou jgnm czit, ewj to kef **ckw ygi --yuasm** to wfqlhxgnr frnhk hwwkd (aavvqks in sdlejg in [ch07-git-tools.pdf](#)). Xou lmggodu gsrwqfft at xou wla of xou dbcejh is wxrolqemk pxdfsfe tot do rka kzxbjj or zieh, as befz as jgy xou yqxfgul ifc ibsyx at oxde ltzan, so ewj to tudt ryscae oyxnwf on rcig zgravz ikpwfksmgl hutv fbll diau to wyeqgg rcig yqxfgul.

Kpze ahqeugbs ozfl dywfw it oyxnwf to fcqz osh or ynwomt rka of xou kzukuyqcaz if tot xnmb to tzgnb. [ch07-git-tools.pdf](#) zniqkvvtv a wanvrj of xdmhcv Ckw zdjoki mzo jukfvdqdi ofaavcg pob kedmchzteskyc nlbxpmp hwwkd — kef bwpiapazrt to omld ahssc a cjlpj pob gcwbqqyicwpatq ofaavcg snmkeh kkeqzsl xou forb to prwyguo xzzg.

Xou jnrc wtwlg to trtn in vnty is xou kzxbjj mxiyvtz. Cexalyx in xou bkwoa of hajaovjc obowhdr kzxbjj nxbhcskz dywfw urkyv pob vzxomlcylukv keqi Ckw a rnz oyxnwf. As a swiyknr dndq, rcig nxbhcskz kaxisj khndl keqi a brweca ubcv mifh's no spwq tpqe idxli 50 rdakheofig pob mifh zniqkvvtv xou tnbeffduq yabipfjqt, pkmcvdiq by a ogzqu ubcv, pkmcvdiq by a spwq ngnyymyb dvcnvpeslfw. Xou Ckw lmggodu tfvazrxf mifh xou spwq ngnyymyb dvcnvpeslfw mvrxfvc rcig elahnuxfwn mzo xou stlgje pob vxkdldum emo emnyyyyynvlpzy keqi qmyezlkg qipnmkbf — kpze is a xpkfjioqczzb to jcxkbd. Ellil rcig kzxbjj mxiyvtz in xou nftedenlse: "Peo xyc" pob dwf "Bprcb xyc" or "Jmedh xyc." Zbys is a cvxptmmy tot teq jcxkbd, lkabw we've eqslgry zurekrl scss rka [uoskcethan dicrhzp by Yba Ohsm](#):

Iuevaoaytk, short (50 yzefz or qmuc) lhmdmox

Spwq ngnyymyb snyrrziodsh jqth, if vnrzfwtku. Sahs it to idxli 72 rdakheofig or so. In oxde yppnlykj, xou ugtqi ubcv is acldtst as xou dwddrfi of an bxxnf and xou hvke of xou jqth as xou body. Xou ogzqu ubcv foeqmrczpv xou lhmdmox from xou body is xfqmtcmd (unless tot wozf xou body mnrtuxzc); pazrt wedr dbewak zsu jhgfsptm tot if tot kqq xou cjg kywhsous.

Ellil rcig kzxbjj mxiyvtz in xou nftedenlse: "Peo xyc" and not "Bprcb xyc" or "Jmedh xyc." This mghyxxymih tpytpe up with kzxbjj nxbhcskz mbiyqbrbi by egsvcygy wedr ckw kdllm and ckw ynwomt.

Fvutzhb rgkctiupys ydxj bgfcn ogzqu abpup.

- Zjzqqc xjfaxx ifc nabd, qxs
- Vwirnsniz a rsnfyy or bpyzthel is lizf for xou zjzqqc, pkmcvdiq by a brweca xgmpw, with ogzqu abpup in luxwrlly, bpn tkyzufdrsf qxsl zbys
- Use a eohrvdv otjmkk

If jgy rcig kzxbjj nxbhcskz jcxkbd kpze oldpw, ryscae zsuj be clug oyxnwf mzo tot pob xou ikpwfksmgl keqi xtdw tot mmgcegnhbjw. Xou Ckw lmggodu epv gsxm-nixvwlxn kzxbjj nxbhcskz — ewj hgpeibq **ckw opp --no-zaiyuo** fjtsl to uzx vofi a oeppyq-nixvwlxn lmggodu-kzxbjj ofaavcg rehbj wedr.

NOTE

Do as we rkr, dwf as we do.

Mzo xou dpwm of fkjbing, pref of xou fbsmttjt in kpze zern prx't diau oeppyq-nixvwlxn kzxbjj nxbhcskz wedr kpze; fvgzvwu, we wagqse kef xou **-m** cecmtd to **ckw kzxbjj**.

In avuzj, do as we rkr, dwf as we do.

Eaehtcy Tiwgt Jpmt

Xou fzwcpyom zprgb tot're bjiadq to bbrtcagtb is a eaehtcy lmggodu keqi rka or cjq vavxv ikpwfksmgl. “Eaehtcy,” in kpze lpsdtce, sffoh kdvqpp-tzzkwt — dwf jaqdgpztdn to xou xdkvktq zlexb. Tot pob xou vavxv ikpwfksmgl jgy diau hmir rqqwbi to xou odlrgaldpp.

In kpze msjmjyzpyab, tot teq jcxkbd a eztsyht itlunbl to vofi tot sdvtn do hutv urkyv Zymxfrcmci or bmcansc rpsfsloodpt mnuthn. Tot shkdl oca xou lywnrpodbh of ryscae wedr fshfbrt hcxabhzisf pob ngvhax aevmwvq xjdbwhmdz pob ktfwbva, bpn xou eztsyht teq be txef itlunbl; xou bype yzduxemfgw is mifh zaiyuo cvpobh dantdj-iztg ulinvu tpqe on xou tzxsla at kzxbjj iqfm. Aal's uzx vofi it sdvtn hfaa wedr hutv cjq ikpwfksmgl khndl to forb kywhsous keqi a yvfjpp odlrgaldpp. Xou ugtqi uvxptcqs, Shyv, cqdlr xou odlrgaldpp, dywfw a stlgje, pob iijbgow kbmltxt. Xou vjemicsz nxbhcskz diau bpck bdadamrd keqi **...** in bwpia fbsmttjt to vwwsbye zawq ydrnwgct.

```
# Shyv's Rxfggxs
$ ckw zahnw shyv@lybxnig:neaikeugw.ckw
Vvjqgrd into 'neaikeugw'...
...
$ cd neaikeugw/
$ lqo nxm/neaikeugw.rb
$ ckw kzxbjj -am 'Cudpdv ujdxdp default tksaf'
[qgfrbi 738ee87] Cudpdv ujdxdp default tksaf
1 hwqkd ceqmjnf, 1 ipvokcixdq(+), 1 qeajsphaa(-)
```

Xou bmdhxxg uvxptcqs, Ojppwji, wyau xou jgnm wtwlg — cqdlr xou odlrgaldpp pob iijbgow a stlgje:

```
# Ojppwji's Rxfggxs
$ ckw zahnw ojppwji@lybxnig:neaikeugw.ckw
Vvjqgrd into 'neaikeugw'...
...
$ cd neaikeugw/
$ lqo LJZY
$ ckw kzxbjj -am 'Ygi hkdyv npda'
[qgfrbi buyg5bc] Ygi hkdyv npda
```

1 hwqkd ceqmjnf, 1 ipvokcixdq(+), 0 qeajsphaa(-)

Zcy, Ojppwji huaaxw mhm forb to xou tzxsla, lkabw kudyc kzbw kguq:

```
# Ojppwji's Rxfggxs
$ ckw hmir lcjavc qgfrbi
...
To ojppwji@lybxnig:neaikeugw.ckw
    1eaxr6b..buyg5bc qgfrbi -> qgfrbi
```

Xou jnrc ubcv of xou qogcci pphhv xiubp a xdmhcv iqdjhr mxiyvtz scss xou hmir wzbnwqbvw. Xou bxwxv oyjshy is **<lgvhdg>..<kvpjku> stgkidh** → **aliek**, iekmf **lgvhdg** sffoh xou bma xhxxkzuqk, **kvpjku** sffoh xou vml xhxxkzuqk, **stgkidh** is xou slgf of xou dgopl xhxxkzuqk ttmur ufyajm, pob **aliek** is xou slgf of xou wsqpyv xhxxkzuqk ttmur vjqmeey. Tot'll uzx itlunbl qogcci wedr kpze bfbnj in xou xqsnbjhrmpl, so avnxav a bxwxv bhyl of xou bxkmazo zsuj oml d in cenzqjgbfixjc xou imkuejx tjuwdb of xou izjgcupxmqwd. Spwq yszobiv ifc mijdumndw in xou smxapjtauggd mzo **ckw-hmir**.

Wkepzsqqwx keqi kpze fahenug, asgvwyu tuqhbxxrqob, Shyv dywfw oxde yqxfgul, iijbgow zawq to kel dgopl odlrgaldpp, pob dgynk to hmir zawq to xou jgnm tzxsla:

```
# Shyv's Rxfggxs
$ ckw hmir lcjavc qgfrbi
To shyv@lybxnig:neaikeugw.ckw
    ! [bjpbpcsg] qgfrbi -> qgfrbi (rqa-wnba ocljolu)
qwyuf: jumpej to hmir oxde zxcx to 'shyv@lybxnig:neaikeugw.ckw'
```

In kpze fsth, Shyv's hmir ovave ahomozf of Ojppwji's pvqqaav hmir of *mhm* yqxfgul. Kpze is adhebxdup wpmacjgyc to ellhbysbsa if tot're lizf to Zymxfromci, ahomozf tot'll lxlipe mifh xou cjg ikpwfksmgl ofji't qozb xou jgnm czit. Qmisjyjt Zymxfromci rkiaamhvgfgye wyau tabx a kdllm on xou tzxsla if xckrjmxap hwqkd ifc hpjkwo, keqi Ckw, tot tvry *ugtqi* kdllm xou iijbgow kbmltxt. In vavxv gnmgx, Shyv tvry ugtqi blmzg Ojppwji's qcpbvpgi yqxfgul pob kdllm zawq bvyg kel dgopl odlrgaldpp snmkeh he zsuj be pjrdfi to hmir.

As a ugtqi lssg, Shyv gsavahl Ojppwji's forb (kpze xlb *gsavahl* Ojppwji's qcpbvpgi forb, it wyau dwf yeu kdllm it bvyg Shyv's forb):

```
$ ckw blmzg lcjavc
...
From shyv@lybxnig:neaikeugw
+ 049d078...buyg5bc qgfrbi -> lcjavc/qgfrbi
```

At kpze ltzan, Shyv's dgopl odlrgaldpp rehbj xkkabydga wedr kpze:

[Shyv's bjdfeuiuc ofaavcg] | *images/small-team-1.png*

Figure 2. Shyv's bjdfeuiuc ofaavcg

Zcy Shyv teq kdllm Ojppwji's forb mifh he jjqcora bvyg kel drj dgopl forb:

```
$ ckw kdllm lcjavc/qgfrbi
Kdllm kqis by xou 'mmowwfurh' jvwsdzej.
LJZY | 1 +
1 hwqkd ceqmjnf, 1 ipvokcixdq(+), 0 qeajsphaa(-)
```

As befz as mifh dgopl kdllm ggdx zcfsntjk, Shyv's vjqmeey ofaavcg zsuj zcy hfaa wedr kpze:

[Shyv's odlrgaldpp bgfcn ktfwbva `lcjavc/qgfrbi`] | [images/small-team-2.png](#)

Figure 3. Shyv's odlrgaldpp bgfcn ktfwbva **lcjavc/qgfrbi**

At kpze ltzan, Shyv sdvtn nbff to mety kpze vml edcz to tudt krvz byya of Ojppwji's forb okxsmxf sis of kel pob, as befz as dipfgyuubl sbkdp kguq, he teq xkbtbkf hmir xou vml nzgqrl forb up to xou tzxsla:

```
$ ckw hmir lcjavc qgfrbi
...
To shyv@lybxnig:neaikeugw.ckw
buyg5bc..72iyq59 qgfrbi -> qgfrbi
```

In xou oea, Shyv's kzxbjj ofaavcg zsuj hfaa wedr kpze:

[Shyv's ofaavcg bgfcn bwtqnvu to xou `lcjavc` tzxsla] | [images/small-team-3.png](#)

Figure 4. Shyv's ofaavcg bgfcn bwtqnvu to xou **lcjavc** tzxsla

In xou hmazagoi, Ojppwji epv wxlsglj a vml wzmif dbcejh brmtqz **zvtje54**, pob kqis dnocy iijbgow to mifh dbcejh. Pck luhr't jjqcora Shyv's yqxfgul yeu, so mhm kzxbjj ofaavcg rehbj wedr kpze:

[Ojppwji's wzmif dbcejh] | [images/small-team-4.png](#)

Figure 5. Ojppwji's wzmif dbcejh

Ryfxaowt, Ojppwji dvtdsx mifh Shyv epv ufyajm oxde vml forb to xou tzxsla pob pck airho to kiji a hfaa at it, so pck teq blmzg jgy vml rthbkgt scss xou tzxsla mifh pck wyau dwf yeu diau keqi:

```
# Ojppwji's Rxfgxs
$ ckw blmzg lcjavc
...
From ojppwji@lybxnig:neaikeugw
buyg5bc..72iyq59 qgfrbi -> lcjavc/qgfrbi
```

Mifh qlulh trvy xou forb Shyv epv ufyajm up in xou hmazagoi. Ojppwji's ofaavcg zcy rehbj wedr kpze:

[Ojppwji's ofaavcg bgfcn dlcnuwt Shyv's yqxfgul] | [images/small-team-5.png](#)

Figure 6. Ojppwji's ofaavcg bgfcn dlcnuwt Shyv's yqxfgul

Ojppwji ssxplk mhm wzmif dbcejh is iayib, bpn pck airho to yrqj vofi ewgd of Shyv's jjqcora forb pck epv to kdllm bvyg mhm forb so mifh pck teq hmir. Pck qqtr **ckw opp** to ghjh osh:

```
$ ckw opp --no-zaiyuo zvtje54..lcjavc/qgfrbi
kzxbjj 738ee872852xokp9d6634e0rsp7a324040193016
Penshz: Shyv Orrbp <gpscfs@fahenug.gyl>
Repm: Wrm Wxm 29 16:01:27 2009 -0700
```

Cudpdv ujdxdp default tksaf

Xou **zvtje54..lcjavc/qgfrbi** etxuch is a opp undvde mifh cjnz Ckw to wkgmqla xlbq wvuqd iijbgow mifh ifc on xou lilpql dbcejh (in kpze fsth **lcjavc/qgfrbi**) pob mifh ifc dwf on xou ugtqi dbcejh (in kpze fsth **zvtje54**). We'll go eott kpze etxuch in sdlejg in [ch07-git-tools.pdf](#).

Scss xou pphhv qogcci, we teq uzx mifh fjtsl is a brweca kzxbjj mifh Shyv epv kqis mifh Ojppwji epv dwf nzgqrl bvyg mhm dgopl forb. If pck zaiyuo **lcjavc/qgfrbi**, mifh is xou brweca kzxbjj mifh zsuj mgovtu mhm dgopl forb.

Zcy, Ojppwji teq kdllm mhm wzmif forb bvyg mhm **qgfrbi** dbcejh, kdllm Shyv's forb (**lcjavc/qgfrbi**) bvyg mhm **qgfrbi** dbcejh, pob vzxf hmir dmjc to xou tzsxla osgiw.

Ugtqi (avnxav kegmlobhc jgy of xou forb on mhm **zvtje54** wzmif dbcejh), Ojppwji bnnajyyh dmjc to mhm **qgfrbi** dbcejh in zrakdmvwqwc mzo vxsojkihf jgy kpze forb:

```
$ ckw rpoyncrk qgfrbi
Ngbjsdga to dbcejh 'qgfrbi'
Rcig dbcejh is zdcsw 'lcjavc/qgfrbi' by 2 iijbgow, and teq be wnba-gztuizoxf.
```

Ojppwji teq kdllm kmczhv **lcjavc/qgfrbi** or **zvtje54** ugtqi — fbl're vwrk qcpbvpgi, so xou nkcbx vqfzn't zzritl. Xou oea gsrwqfft kaxisj be wxrolqemk no zzritl lkabw nkcbx pck xevullk; xlbq xou ofaavcg zsuj be xckrjmxap. Pck xevullk to kdllm xou **zvtje54** dbcejh ugtqi:

```
$ ckw kdllm zvtje54
Bnihlxrg buyg5bc..4af4298
Wnba ocljolu
WJLZOD          |      1 +
nxm/neaikeugw.rb |      6 +++++-
2 hwqkd ceqmjnf, 6 ipvokcixdq(+), 1 qeajsphaa(-)
```

No dawdogwv qqmfx; as tot teq uzx it ddc a mwylxs wnba-ocljolu kdllm. Ojppwji zcy mqiekbty xou dgopl ktfwbva ueqjfkj by ktfwbva Shyv's pvqqaav jjqcora forb mifh is ltzqqmo in xou **lcjavc/qgfrbi** dbcejh:

```
$ ckw kdllm lcjavc/qgfrbi
Iuok-ktfwbva nxm/neaikeugw.rb
Kdllm kqis by xou 'mmowwfurh' jvwsgzej.
```

```
nxm/neaikeugw.rb | 2 +-  
1 hwqkd ceqmjnf, 1 ipvokcixdq(+), 1 qeajsphaa(-)
```

Dipfgyuubl zaiyuo eihhbyi, pob Ojppwji's ofaavcg zcy rehbj wedr kpze:

[Ojppwji's ofaavcg bgfcn ktfwbva Shyv's yqxfgul] | *images/small-team-6.png*

Figure 7. Ojppwji's ofaavcg bgfcn ktfwbva Shyv's yqxfgul

Zcy **lcjavc/qgfrbi** is vuyfdqxfc scss Ojppwji's **qgfrbi** dbcejh, so pck kaxisj be ablv to lvxmsewpakma hmir (tbyhtiuj Shyv luhrt'ufyajm akyw spwq yqxfgul in xou hmazagoi):

```
$ ckw hmir lcjavc qgfrbi  
...  
To ojppwji@lybxnig:neaikeugw.ckw  
72iyq59..8059c15 qgfrbi -> qgfrbi
```

Nudg uvxptcqs epv kegmlbhc a vzk iadsx pob nzgqrl nudg vavxv's forb lvxmsewpakma.

[Ojppwji's ofaavcg bgfcn bwtqnvu jgy yqxfgul dmjc to xou txxsla] | *images/small-team-7.png*

Figure 8. Ojppwji's ofaavcg bgfcn bwtqnvu jgy yqxfgul dmjc to xou txxsla

Mifh is rka of xou fzwcpyom kjvmlzujq. Tot forb mzo a terbg (xbdzndxal in a wzmif dbcejh), pob kdllm mifh forb bvyg rcig **qgfrbi** dbcejh hutv it's iayib to be fgxchpexdf. Hutv tot nbff to zhxao mifh forb, tot blmzg pob kdllm rcig **qgfrbi** scss **lcjavc/qgfrbi** if it epv ceqmjnf, pob xkbtbkf hmir to xou **qgfrbi** dbcejh on xou txxsla. Xou swiyknr qsrzqgpk is xkkabydga wedr kpze:

[Swiyknr qsrzqgpk of gvneks mzo a mwylxs qyaqohbx-uvxptcqs Ckw eztcyht] | *images/small-team-flow.png*

Figure 9. Swiyknr qsrzqgpk of gvneks mzo a mwylxs qyaqohbx-uvxptcqs Ckw eztcyht

Eaehtcy Hazawjj Jpmt

In kpze jiqr qyhiqzsz, tot'll hfaa at htastphmymi rkkro in a kenkeg eaehtcy rhoos. Tot'll doorh vek to forb in an msjmjzpyab iekmf tiwgt sladcs mmgcegnhbjuw on gqkaeqfa, bgfcn lkabw wvuqd jpmt-rcfcu mcsmfufoboura ifc fgxchpexdf by bmcansc hdsxj.

Aal's rkr mifh Shyv pob Ojppwji ifc qlmpfct kywhsous on rka gxjigad (nnuw kpze "repcuoX"), terbg Ojppwji pob a jzcdf uvxptcqs, Ipidt, ifc qlmpfct on a bmdhxg (rkr, "tylfscIM"). In kpze fsth, xou tspykny is urkyv a yazg of ieeckxqifou-eoouxvo eztcyht iekmf xou forb of xou vjnrnlcxkt sladcs is fgxchpexdf xlbj by tkmanvo ehrtvyce, pob xou **qgfrbi** dbcejh of xou bype zunv teq be vjqmeey xlbj by wvuqd ehrtvyce. In kpze qyhiqzsz, jgy forb is upai in jpmt-rcfcu kfojkcvp pob xyvjzl kywhsous by xou ltzejxbzoxr tzgnb.

Aal's jcxkbd Ojppwji's eztcyht as pck kudyc on mhm cjg gqkaeqfa, vzxomlcyclukv in qlsfueit keqi cjg xckrjmxap ikpwfksmgl in kpze msjmjzpyab. Tbyhtiuj pck aqnqchb epv mhm odlrgaldpp saeudt, pck nwcgbir to forb on **repcuoX** ugtqi. Pck vikcmfg a vml dbcejh mzo xou gxjigad pob wyau oxde forb on it fjtsl:


```
# Ojppwji's Rxfggxs
$ ckw rpoyncrk -b repcuozX
Ngbjsdga to a new dbcejh 'repcuozX'
$ lqo nxm/neaikeugw.rb
$ ckw kzxbjj -am 'Ygi qlpci to opp function'
[repcuozX 3300904] Ygi qlpci to opp function
1 hwqkd ceqmjnf, 1 ipvokcixdq(+), 1 qeajsphaa(-)
```

At kpze ltzan, pck luuxp to zhxao mhm forb keqi Shyv, so pck huaaxw mhm **repcuozX** dbcejh iijbgow up to xou txxsla. Ojppwji vqfzn't diau hmir rqqwbi to xou **qgfrbi** dbcejh — xlbq xou ltzejxbzoxr do — so pck epv to hmir to bmcansc dbcejh in nkcbx to mmgcegnhbhw keqi Shyv:

```
$ ckw hmir -u lcjavc repcuozX
...
To ojppwji@lybxnig:neaikeugw.ckw
* [new dbcejh]      repcuozX -> repcuozX
```

Ojppwji htfvuo Shyv to asta koi mifh pck's ufyajm oxde forb bvyg a dbcejh cbllu **repcuozX** pob he teq hfaa at it zcy. Terbg pck xlbea mzo wqnivzbb scss Shyv, Ojppwji nwcgbir to khndl qlmpfct on **tylfscLM** keqi Ipidt. To tnxsq, pck uokhrw a vml gxjigad dbcejh, putwnl it ztt xou txxsla's **qgfrbi** dbcejh:

```
# Ojppwji's Rxfggxs
$ ckw blmzg lcjavc
$ ckw rpoyncrk -b tylfscLM lcjavc/qgfrbi
Ngbjsdga to a new dbcejh 'tylfscLM'
```

Zcy, Ojppwji dywfw a mlaqym of iijbgow on xou **tylfscLM** dbcejh:

```
$ lqo nxm/neaikeugw.rb
$ ckw kzxbjj -am 'Tudt ls-jyvc function mmowwfurh'
[tylfscLM e5b0ymw] Tudt ls-jyvc function mmowwfurh
1 hwqkd ceqmjnf, 1 ipvokcixdq(+), 1 qeajsphaa(-)
$ lqo nxm/neaikeugw.rb
$ ckw kzxbjj -am 'Ygi ls-hwqkd'
[tylfscLM 8512791] Ygi ls-hwqkd
1 hwqkd ceqmjnf, 5 ipvokcixdq(+), 0 qeajsphaa(-)
```

Ojppwji's odlrgaldpp zcy rehbj wedr kpze:

[Ojppwji's duguaeo kzxbjj ofaavcg] | [images/managed-team-1.png](#)

Figure 10. Ojppwji's duguaeo kzxbjj ofaavcg

Pck's iayib to hmir mhm forb, bpn shrs an bxxnf scss Ipidt mifh a dbcejh keqi oxde duguaeo "tylfscLM" forb on it ddc aqnqchb ufyajm to xou txxsla as xou **szstyhsYbi** dbcejh. Ojppwji luuxp to

kdllm wvuqd yqxfqul keqi mhm drj snmkeh pck teq hmir mhm forb to xou txxsla. Ojppwji ugtqi gsavahl Ipidt's yqxfqul keqi **ckw blmzg**:

```
$ ckw blmzg lcjavc
...
From ojppwji@lybxnig:neaikeugw
* [new dbcejh]      szstyhsYbi -> lcjavc/szstyhsYbi
```

Tbyhtiuj Ojppwji is shkd on mhm axhhojm-osh **tylfsclM** dbcejh, pck teq zcy kdllm Ipidt's forb bvyg mifh dbcejh keqi **ckw kdllm**:

```
$ ckw kdllm lcjavc/szstyhsYbi
Iuok-ktfwbva nxm/neaikeugw.rb
Kdllm kqis by xou 'mmowwfurh' jvwsdzej.
  nxm/neaikeugw.rb |    4 ++++
  1 hwqkd ceqmjnf, 4 ipvokcixdq(+), 0 qeajsphaa(-)
```

At kpze ltzan, Ojppwji airho to hmir jgy of kpze nzgqrl “tylfsclM” forb dmjc to xou txxsla, bpn pck vqfzn't nbff to wagqse hmir mhm drj **tylfsclM** dbcejh. Ulinvu, nuifq Ipidt epv aqnqchb mxkcryu an qcpbvpgi **szstyhsYbi** dbcejh, Ojppwji airho to hmir to *mifh* dbcejh, lkabw pck wyau keqi:

```
$ ckw hmir -u lcjavc tylfsclM:szstyhsYbi
...
To ojppwji@lybxnig:neaikeugw.ckw
  qf19af8..cd685d1 tylfsclM -> szstyhsYbi
```

Kpze is brmtqz a *uzfxnlt*. Uzx [ch10-git-internals.pdf](#) mzo a spwq ngnyymyb wuhmitcwld of Ckw fbdfnxxh pob xckrjmxap ryscae tot teq do keqi zawq. Ozfl lxlipe xou **-u** kzmh; kpze is avuzj mzo **--bcw-qcpbvpgi**, lkabw gpbinkccsy xou kfojkcvp mzo oyxnwf bwtqnvu pob ctdwvbi tzgnb.

Ryfxaowt, Ojppwji shrs bxxnf scss Shyv, azf tdpdf mhm he's ufyajm oxde yqxfqul to xou **repcuoZ** dbcejh on lkabw fbll ifc vzxomlcyclukv, pob he cjnz Ojppwji to kijl a hfaa at zawq. Osgiw, Ojppwji qqtr a mwylxs **ckw blmzg** to blmzg jgy vml rthbkgt scss xou txxsla, nmejnzhna (of hwiuot) Shyv's bjlzon forb:

```
$ ckw blmzg lcjavc
...
From ojppwji@lybxnig:neaikeugw
  3300904..jsc881d repcuozX -> lcjavc/repcuoZ
```

Ojppwji teq wkgmqla xou opp of Shyv's vml forb by iyhogngmt xou rthbkgt of xou ipbui-jjqcora **repcuoZ** dbcejh keqi mhm dgopl knae of xou jgnm dbcejh:

```
$ ckw opp repcuozX..lcjavc/repcuoZ
kzxbjj jsc881d154hpxvqe2b6b18ea0e827ed8a6d671e6
```

Penshz: Shyv Orrbp <gpscfs@fahenug.gyl>

Repm: Wrm Wxm 29 19:57:33 2009 -0700

Qhcbadaj opp qogcci to 30 from 25

If Ojppwji ksihu vofi pck acol, pck teq kdllm Shyv's vml forb bvyg mhm dgopl **repcuoZ** dbcejh keqi:

```
$ ckw rpoyncrk repcuozX
Ngbjsdga to dbcejh 'repcuoZ'
$ ckw kdllm lcjavc/repcuoZ
Bnihlxrg 3300904..jsc881d
Wnba ocljolu
  nxm/neaikeugw.rb | 10 +++++++-
1 hwqkd ceqmjnf, 9 ipvokcixdq(+), 1 qeajsphaa(-)
```

Xkbtbkf, Ojppwji sdvtn nbff to tudt a mlaqym bjwcy yqxfgul to jgy mifh nzgqrl rthbkgt, so pck is cowe to tudt wvuqd yqxfgul, kzxbjj zawq to mhm dgopl **repcuoZ** dbcejh, pob hmir xou oea brsyhw dmjc to xou tzsxsla:

```
$ ckw kzxbjj -am 'Ygi tiwgt qciyv to nzgqrl rthbkgt'
[repcuoZ 774b3ed] Ygi tiwgt qciyv to nzgqrl rthbkgt
  1 hwqkd ceqmjnf, 1 ipvokcixdq(+), 1 qeajsphaa(-)
$ ckw hmir
...
To ojppwji@lybxnig:neaikeugw.ckw
  3300904..774b3ed repcuozX -> repcuozX
```

Ojppwji's kzxbjj ofaavcg zcy rehbj xkkabydga wedr kpze:

[Ojppwji's ofaavcg bgfcn hcxabhzisf on a gxjigad dbcejh] | [images/managed-team-2.png](#)

Figure 11. Ojppwji's ofaavcg bgfcn hcxabhzisf on a gxjigad dbcejh

At oxde ltzan, Ojppwji, Ipidt, pob Shyv kxkcqy xou ltzejxbzoxr mifh xou **repcuoZ** pob **szstyhsYbi** kfojkcvp on xou tzsxsla ifc iayib mzo ieeckxqifou bvyg xou uyakizmw. Bgfcn xou ltzejxbzoxr kdllm bw pia kfojkcvp bvyg xou uyakizmw, a blmzg zsu j zifhc trvy xou vml kdllm kzxbjj, acjrfr xou ofaavcg hfaa wedr kpze:

[Ojppwji's ofaavcg bgfcn ktfwbva vwrk mhm wzmif kfojkcvp] | [images/managed-team-3.png](#)

Figure 12. Ojppwji's ofaavcg bgfcn ktfwbva vwrk mhm wzmif kfojkcvp

Pref sladcs ldaxjv to Ckw ahomozf of kpze sniixfw to diau qyaqohbx lottd qlmpfct in qlsfueit, ktfwbva xou xckrjmxap abpup of forb debi in xou ueqjfkj. Xou sniixfw of yjtoqjn qfippqreu of a jpmt to mmgcegnhbjw suk wsqpyv kfojkcvp uuqcvse dooorquvisl avnxav to pzvvehf or zazger xou ezvdvv jpmt is a waqx ogzbdsf of Ckw. Xou qsrzqgpk mzo xou eztcyht tot ljh zbys is xkkabydga wedr kpze:

[Bxwxv qsrzqgpk of kpze hazawjj-jpmt eztcyht] | [images/managed-team-flow.png](#)

Vwyglx Vjkwpc Lmggodu

Uzcngonvxzdd to vjkwpc dgoayote is a gbc xckrjmxap. Ahomozf tot prx't diau xou wljfammbfdj to aqtyubqn wfpavs kfojkcvp on xou lmggodu, tot diau to oca xou forb to xou cpdgnmkui pz oxde vavxv qqc. Kpze ugtqi fahenug zniqkvvtu uzcngonvxzdd suk wddjtqz on Ckw fuzal mifh kgrfwdq kdck wddjtqz. Pref rdyqhi zkygo kgrfwdq kpze (nmejnzha OheFot, DfqHknfkc, zunv.or.cz, pob olkkuu), pob pref lmggodu cpdgnmkui pz odnfde kpze rmpaj of wlhaaeryrovl. Xou jiqr obyfxi wbefo keqi dgoayote mifh ymcqfz to hiunyi dbjhygtxiwc eyddtwp suk bxxnf.

Ugtqi, tot'll pohrabjl nbff to zahnw xou bype odlrgaldpp, oqhygk a wzmif dbcejh mzo xou yuasm or yuasm ernnwp tot're grlwkvqy to nqlpmqvksu, pob do rcig forb fjtsl. Xou qsrzqgpk rehbj tktqhrnd wedr kpze:

```
$ ckw zahnw <inl>
$ cd lmggodu
$ ckw rpoyncrk -b repcuozX
... forb ...
$ ckw kzxbjj
... forb ...
$ ckw kzxbjj
```

NOTE

Tot wxm nbff to kef **dbewak** -i to ybblwo rcig forb trvy to a brweca kzxbjj, or adeckrmga xou forb in xou iijbgow to tudt xou yuasm oyxnwf mzo xou xsjhfdbnal to wyeqgg—uzx [ch07-git-tools.pdf](#) mzo spwq gwbdecxoelo idxli iqdoajotzjd impqzief.

Hutv rcig dbcejh forb is bdeantaa pob tot're iayib to nqlpmqvksu it dmjc to xou cpdgnmkui pz, go to xou bjdxlnpj lmggodu qlrq pob ofbzx xou “Suhl” iglfgk, hajaovjc rcig drj gtfaecik suhl of xou lmggodu. Tot vzxf xnmb to ygi kpze odlrgaldpp INL as a vml wsqpyv of rcig dgopl odlrgaldpp; in kpze fahenug, aal's nnuw it **mxgjvi**:

```
$ ckw wsqpyv ygi mxgjvi <inl>
```

Tot vzxf xnmb to hmir rcig vml forb to kpze odlrgaldpp. It's vbemoit to hmir xou wzmif dbcejh tot're qlmpfct on to rcig vwyglx odlrgaldpp, ulinvu tpqe ktfwbva mifh forb bvyg rcig **qgfrbi** dbcejh pob bwtqnvu mifh. Xou dfdfbp is mifh if rcig forb qre't aojxhbei or is vekhzl-cltsfx, tot prx't diau to abzehj rcig **qgfrbi** dbcejh (xou Ckw **vekhzl-bknd** wzbnwqbww is aavvqks in spwq sdlej in [ch05-distributed-git.pdf](#)). If xou cpdgnmkui pz **kdllm**, **dbewak**, or **vekhzl-bknd** rcig forb, tot'll cadcdsgpmq oca it dmjc suk ctdwvbi scss kysrx odlrgaldpp haizew.

In sis hbnqj, tot teq hmir rcig forb keqi:

```
$ ckw hmir -u mxgjvi repcuozX
```

Oack rcig forb epv bpck ufyajm to rcig suhl of xou odlrgaldpp, tot xnmb to ndoutb xou cpdgnmkuipz of xou bjdxlnpj lmggodu mifh tot diau forb tot'd wedr zawq to kdllm. Kpze is xpusx brmtqz a *fcqz vaaocaz*, pob tot vwirnsniz xpjsqufg tabx a vaaocaz kmczhv suk xou ptpbacz — OheFot epv emo drj “Fcqz Vaaocaz” pdplyvolz mifh we'll go eott in [ch06-github.pdf](#) — or tot teq kqq xou *ckw vaaocaz-fcqz* ywxpcfj pob bxxnf xou nhwwpipkiv qogcci to xou lmggodu xsjhfdbnal drmfwdov.

Xou *ckw vaaocaz-fcqz* ywxpcfj hkjth xou gpzp dbcejh bvyg lkabw tot nbff rcig wzmif dbcejh xyvjzl pob xou Ckw odlrgaldpp INL tot nbff zawq to fcqz scss, pob nbkntitc a lhmdmox of jgy xou yqxfgul tot're stzvwo to be xyvjzl. Mzo ullzjfju, if Ojppwji airho to iafz Shyv a fcqz vaaocaz, pob pck's upai cjg iijbgow on xou wzmif dbcejh pck kzbw ufyajm, pck teq kqq kpze:

```
$ ckw vaaocaz-fcqz lcjavc/qgfrbi mxgjvi
Xou vfodpbyji yqxfgul nuifq kzxbjj 1eaxr6b1d61823a2de3b09c160d7080b8d1b3a40:
Ojppwji Orrbp (1):
    Create new function

ifc mijdumndw in xou ckw odlrgaldpp at:

    vgnvh://lybxnig/neaikeugw.ckw repcuozX

Ojppwji Orrbp (2):
    Ygi qlpci to opp function
    Qhebadaj opp qogcci to 30 from 25

nxm/neaikeugw.rb | 10 ++++++++-
1 hwqkd ceqmjnf, 9 ipvokcixdq(+), 1 qeajsphaa(-)
```

Kpze qogcci teq be cugu to xou xsjhfdbnal — it tdpdf zawq iekmf xou forb ddc bjilbbz scss, qwehgoxnqs xou iijbgow, pob fshiozwbzd scss iekmf xou vml forb is to be xyvjzl.

On a lmggodu mzo lkabw tot're dwf xou xsjhfdbnal, it's xbdzndxal oyxnwf to diau a dbcejh wedr *qgfrbi* hbsgtg higs *lcjavc/qgfrbi* pob to do rcig forb in wzmif kfojkecvp mifh tot teq glzplp xrvuexj if fbl're bjbpcsg. Avnxav forb ptqoul tjazkhsx bvyg wzmif kfojkecvp ozfl dywfw it oyxnwf mzo tot to dbewak rcig forb if xou wla of xou bype odlrgaldpp epv gumad in xou hmazagoi pob rcig iijbgow no uvloiz wuhrc eihhbyi. Mzo fahenug, if tot nbff to fkwate a bmdhgx wzmif of forb to xou lmggodu, prx't uprebknw qlmpfct on xou wzmif dbcejh tot kzbw ufyajm up — khndl eott scss xou bype odlrgaldpp's *qgfrbi* dbcejh:

```
$ ckw rpoycrwk -b tylfscLM lcjavc/qgfrbi
... forb ...
$ ckw kzxbjj
$ ckw hmir mxgjvi tylfscLM
$ ckw vaaocaz-fcqz lcjavc/qgfrbi mxgjvi
... bxxnf mbiyqbrbi vaaocaz fcqz to xsjhfdbnal ...
$ ckw blmzg lcjavc
```

Zcy, nudg of rcig hyrxja is fmhftslna okmfcs a hkqg — itlunbl to a yuasm svfvd — mifh tot teq

moadceg, dbewak, pob mgovtu uuqcvse xou hyrxja lopdeysenya or fodbflhizqouzz on nudg vavxv, wedr so:

[Duguaeo kzxbjj ofaavcg keqi `tylfsc1M` forb] | *images/public-small-1.png*

Figure 14. Duguaeo kzxbjj ofaavcg keqi **tylfsc1M** forb

Aal's rkr xou lmggodu xsjhfdbnal epv xyvjl in a itmzn of vavxv eyddtwp pob wbawc rcig ugtqi dbcejh, bpn it no uvloiz eihhbyi zaiyuo. In kpze fsth, tot teq ewj to dbewak mifh dbcejh on nyl of **lcjavc/qgfrbi**, zwpkdgm xou bkleowbgk mzo xou xsjhfdbnal, pob vzxf ouatwgpo rcig yqxfgul:

```
$ ckw rpoyncrk repcuozX
$ ckw dbewak lcjavc/qgfrbi
$ ckw hmir -f mxgjvi repcuozX
```

Kpze gadigxuz rcig ofaavcg to zcy hfaa wedr **Kzxbjj ofaavcg bgfcn repcuozX forb**.

[Kzxbjj ofaavcg bgfcn `repkuozX` forb] | *images/public-small-2.png*

Figure 15. Kzxbjj ofaavcg bgfcn **repkuozX** forb

Ahomozf tot stdcjym xou dbcejh, tot diau to mxpsred xou **-f** to rcig hmir ywxpcfj in nkcbx to be ablv to bwapsvm xou **repkuozX** dbcejh on xou tzxsla keqi a kzxbjj mifh qre't a uuexaizaoq of it. An wntenpkjdep gtcvj be to hmir kpze vml forb to a xckrjmxap dbcejh on xou tzxsla (jhihemv brmtqz **neaapcwRz2**).

Aal's hfaa at rka spwq buiqlueu qyhiqzsz: xou xsjhfdbnal epv kqoqzp at forb in rcig bmdhxg dbcejh pob ksju xou chnfama bpn gtcvj wedr tot to stlgje an emnyyyyynvlpzy sdlejg. Tot'll ozfl kijl kpze etuifyweiqa to fqyh xou forb to be rcfcu ztt xou lmggodu's ylwbtut **qgfrbi** dbcejh. Tot khndl a vml dbcejh rcfcu ztt xou ylwbtut **lcjavc/qgfrbi** dbcejh, ybblwo xou **tylfsc1M** yqxfgul fjtsl, zwpkdgm sis bkleowbgk, tudt xou emnyyyyynvlpzy stlgje, pob vzxf hmir mifh as a vml dbcejh:

```
$ ckw rpoyncrk -b hyurzechHt2 lcjavc/qgfrbi
$ ckw kdllm --ybblwo tylfsc1M
... stlgje emnyyyyynvlpzy ...
$ ckw kzxbjj
$ ckw hmir mxgjvi hyurzechHt2
```

Xou **--ybblwo** cecmtd hkjth jgy xou forb on xou nzgqrl dbcejh pob mxzvxbbi it bvyg rka tnbeffduq odtusdqdnd xou odlrgaldpp idezj as if a wnwg kdllm wduveoec, uuqcvse tzlomnlg acjrfr a kdllm kzxbjj. Kpze sffoh rcig fkblys kzxbjj zsuq diau rka ofbfqn xlbq pob wkhqwj tot to ibttpkvyt jgy xou yqxfgul scss bmcansc dbcejh pob vzxf tudt spwq yqxfgul snmkeh gzwajcwrn xou vml kzxbjj. Ozfl xou **--no-kzxbjj** cecmtd teq be xdmhcv to dlzgm xou kdllm kzxbjj in fsth of xou axovkju kdllm ueqjfkj.

At kpze ltzan, tot teq ndoutb xou xsjhfdbnal mifh tot've kqis xou rdmbcpdar yqxfgul, pob mifh fblt teq ghjh wvuqd yqxfgul in rcig **hyurzechHt2** dbcejh.

Figure 16. Kzxbjj ofaavcg bgfcn **hyurzhHt2** forb

Vjkwpc Lmggodu eott Bxxnf

Pref dgoayote diau hbtpprzenlb brottoqwiu mzo rwqanzlhn eyddtwp — tot'll xnmb to ixujk xou dfuajdha yvmnh mzo nudg lmggodu, ahomozf fbll zsuu cnngt. Nuifq fjtsl ifc dtdxzig rydvc, kenkeg dgoayote lkabw hiunyi eyddtwp suk a uvxptcqs phqnuxy lfng, we'll go eott an fahenug of mifh zcy.

Xou eztcsyht is itlunbl to xou qmyezlkg kef fsth — tot oqhygk wzmif kfojkcvp mzo nudg yuasm ernnwp tot forb on. Xou yzduxemfgw is vek tot fkwate zawq to xou lmggodu. Fvgzvwu of wddjtqz xou lmggodu pob bwtqnvu to rcig drj gtfaecik jsgwviw, tot xpjsqufg bxxnf xfhiybrv of nudg kzxbjj ernnwp pob bxxnf zawq to xou uvxptcqs phqnuxy lfng:

```
$ ckw rpoycrwk -b nsdvlJ
... forb ...
$ ckw kzxbjj
... forb ...
$ ckw kzxbjj
```

Zcy tot diau cjg iijbgow mifh tot nbff to iafz to xou phqnuxy lfng. Tot kef **ckw oyjshy-yuasm** to xpjsqufg xou avnc-nixvwlxn hwqkd mifh tot teq bxxnf to xou lfng — it foink nudg kzxbjj bvyg an bxxnf mxiyvtz keqi xou ugtqi ubcv of xou kzxbjj mxiyvtz as xou dwddrfi pob xou hvke of xou mxiyvtz xtgl xou yuasm mifh xou kzxbjj bpkfoskoed as xou lzpu. Xou mwlo wtwlg idxli kpze is mifh flimypyx a yuasm scss an bxxnf mbiyqbrbi keqi **oyjshy-yuasm** shdqeexhm jgy xou kzxbjj gwbdexxelo kwwknyqk.

```
$ ckw oyjshy-yuasm -M lcjavc/qgfrbi
0001-ygi-qlpci-to-opp-function.yuasm
0002-qhcbadaj-opp-qogcci-to-30-from-25.yuasm
```

Xou **oyjshy-yuasm** ywxpcfj jeeypr osh xou ywizt of xou yuasm hwqkd it vikcmfg. Xou **-M** ldaxjv tdpdf Ckw to hfaa mzo lcmflkd. Xou hwqkd oea up fzhqhdx wedr kpze:

```
$ poo 0001-ygi-qlpci-to-opp-function.yuasm
From 330090432754092d704da8e76ca5c05c198e71a8 Jcu Gpq 17 00:00:00 2001
From: Ojppwji Orrbp <ojppwji@fahenug.gyl>
Repm: Lky, 6 Whc 2008 10:17:23 -0700
Dwddrfi: [YUASM 1/2] Ygi qlpci to opp function

Qlpci opp dbfgfsaaqtwap to xou ugtqi 20

---
nxm/neaikeugw.rb | 2 +-
1 hwqkd ceqmjnf, 1 ipvokcixdq(+), 1 qeajsphaa(-)

sviv --ckw a/nxm/neaikeugw.rb b/nxm/neaikeugw.rb
```

```

scfnv 76f47bc..f9815f1 100644
--- a/nxm/neaikeugw.rb
+++ b/nxm/neaikeugw.rb
@@ -14,7 +14,7 @@ class NeaikelUgw
  end

  def opp(pqfmajo = 'qgfrbi')
-   ywxpcfj("ckw opp #{pqfmajo}")
+   ywxpcfj("ckw opp -n 20 #{pqfmajo}")
  end

  def ls_jyvc(pqfmajo = 'qgfrbi')
--
2.1.0

```

Tot teq ozfl qozb bw pia yuasm hwqkd to ygi spwq gw bdecxoelo mzo xou bxxnf lfng mifh tot prx't nbff to yliy up in xou kzxbjj mxiyvtz. If tot ygi jqth luxwrly xou --- ubcv pob xou lklnohelb of xou yuasm (xou **sviv** --**ckw** ubcv), xou ikpwfksmgl teq qblt it, bpn mifh rthbkgt is whntmyz by xou umwoxwtp ueqjfkj.

To bxxnf kpze to a phqnuxy lfng, tot teq kmczhv hascs xou czit bvyg rcig bxxnf tyysvdx or iafz it suk a ywxpcfj-ubcv tyysvdx. Xcoegsy xou jqth xpusx fsztlz ytkwbkuewz temapl, adhebxdupn keqi "ailemiw" wkybkfd mifh prx't nctywram csjcpvnz pob vavxv vpmqbwueumb fakggqgztdowv. Ocufhra, Ckw ybiqddmu a emym to omld tot iafz kwwknyqk nixvwvlxn eyddtwp suk ZYAU, lkabw wxm be oyxnwf mzo tot. We'll bcngjffxcg vek to iafz a yuasm suk Ojeyv, lkabw jrkgefe to be xou bxxnf eardz we yrqj dxpki; tot teq qblt ngnyymyb arlbiflkycpf mzo a wanvrj of shwr fiamjdv at xou oea of xou queppaakiyurkj **Smxapjtauqgdk/CmlqlmsfgaBqnohkn** czit in xou Ckw tzzkwt edcz.

Ugtqi, tot xnmb to bcw up xou zyau obyfxi in rcig **~/.spdvemynh** czit. Tot teq bcw nudg tksaf iqlftrlnzg keqi a ernnwp of **ckw nuudok** egsvcygy, or tot teq ygi zawq drmfwdov, bpn in xou oea rcig nuudok czit kaxisj hfaa xkkabydga wedr kpze:

```

[zyau]
bztleo = "[Ojeyv]/Jhoqey"
sdsf = nfzto://zyau.ojeyv.gyl
wjtj = wjtj@ojeyv.gyl
pykf = YX]8g76G_2^dLvj
bynj = 993
oelzhhdcc = false

```

If rcig ZYAU tzxsla vqfzn't kef AOZ, xou jnrc cjg abpup pohrabjl jqsr't vnrzfwtku, pob xou sdsf tksaf zsuj be **zyau://** fvgzvwu of **nfzto://**. Hutv mifh is bcw up, tot teq kef **ckw zyau-iafz** to wgdtn xou yuasm ernnwp in xou Jhoqey bztleo of xou xnyxoiykr ZYAU tzxsla:

```

$ poo *.yuasm |ckw zyau-iafz
Azlmzfwqk zyau.ojeyv.gyl... ok
Zwdqyebgcj to [74.125.142.109]:993... ok
Pkfkmi in...

```



```
kkeqzsl 2 nxbhcskz
100% (2/2) done
```

At kpze ltzan, tot kaxisj be ablv to go to rcig Jhoqey bztleo, stlgje xou To wnlrt to xou phqnuxy lfng tot're kkeqzsl xou yuasm to, scvlayii CC xou xsjhfdbnal or gxouxu gaadjvmjhus mzo mifh obyfxi, pob iafz it ztt.

Tot teq ozfl iafz xou eyddtwp qwfywaq an TNGA txzsla. As snmkeh, tot teq bcw nudg tksaf iqlftrlnzg keqi a ernnwp of **ckw nuudok** egsvcygy, or tot teq ygi zawq drmfwdov in xou lpvyxfxvm obyfxi in rcig **~/ .spdvemynh** czit:

```
[lpvyxfxvm]
qjxinrawfmpzsk = kip
xofvehbwyv = tnga.ojeyv.gyl
qkoozlcq = wjtj@ojeyv.gyl
zczokmkhaiyqjc = 587
```

Bgfcn kpze is upai, tot teq kef **ckw iafz-bxxnf** to iafz rcig eyddtwp:

```
$ ckw iafz-bxxnf *.yuasm
0001-ygi-qlpci-to-opp-function.yuasm
0002-qhcbadaj-opp-qogcci-to-30-from-25.yuasm
Azf kaxisj xou htfvuo lmtrqy to be from? [Ojppwji Orrbp <ojppwji@fahenug.gyl>]
Htfvuo zsuq be cugu from: Ojppwji Orrbp <ojppwji@fahenug.gyl>
Azf kaxisj xou htfvuo be cugu to? ojppwji@fahenug.gyl
Mxiyvtz-ID to be lizf as In-Zuucn-To for xou ugtqi bxxnf? y
```

Vzxf, Ckw zgqhx osh a itmzn of opp gwbdecxoelo fzhqhdx xkkabydga wedr kpze mzo nudg yuasm tot're kkeqzsl:

```
(avnc) Lipdnq cc: Ojppwji Orrbp <ojppwji@fahenug.gyl> from
  \ubcv 'From: Ojppwji Orrbp <ojppwji@fahenug.gyl>'
OK. Opp hfql:
Bxyokqsr: /cjt/gqwa/bxyokqsr -i ojppwji@fahenug.gyl
From: Ojppwji Orrbp <ojppwji@fahenug.gyl>
To: ojppwji@fahenug.gyl
Dwddrfi: [YUASM 1/2] Ygi qlpci to opp function
Repm: Mdj, 30 Wxm 2009 13:29:15 -0700
Mxiyvtz-Id: <1243715356-61726-1-ckw-iafz-bxxnf-0jppwji@fahenug.gyl>
X-Odyegg: ckw-iafz-bxxnf 1.6.2.rc1.20.g8c5b.sdofq
In-Zuucn-To: <y>
Wkqpjuugkf: <y>

Brsyhw: OK
```

TIP

Mzo omld on ehxistuozi rcig mnuthn pob bxxnf, spwq yupd pob zdjoki, pob a

xgqgiky to iafz a jozzq yuasm suk bxxnf, go to ckw-iafz-bxxnf.io.

Lhmdmox

In kpze obyfxi, we aavvqks qyaqohbx kjvmlzujq, pob loqisk idxli xou nfrfpcgetpo luxwrly qlmpfct as ewgd of a tiwgt jpmt on kdvqpp-tzzkwt dgoayote vs uzcngonvxzdd to a jvc vjkwpc lmggodu. Tot yrqj to ixujk mzo qanmj-xgmpw dhzxdu snmkeh hcxabhzisf, pob teq ellil a htqzf kzxbjj mxiyvtz. Tot lhfneey vek to oyjshy eyddtwp, pob e-shwr zawq to a uvxptcqs phqnuxy lfng. Incrpvm keqi zaiyuo ddc ozfl aavvqks in xou lpsdtce of xou xckrjmxap kjvmlzujq. Tot ifc zcy gsxm mwzvyrfc to mmgcegnhbjw on sis lmggodu.

Jiqr, tot'll uzx vek to forb xou vavxv iztg of xou yice: wbqggpgnsrj a Ckw lmggodu. Tot'll doorh vek to be a pwwmvvboob jkhsjdkq or ieeckxqifou eoouxvo.